Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class:TYIT	Batch:1
Date of Assignment:04/08/20206	Date/Time of Submission:04/08/2026

a. Develop a PWA that works in offline mode using Service Worker.

Code :

Servise-worker.js

```
self.addEventListener("install", e => {
    e.waitUntil(
        caches.open("v1").then(c =>
            c.add("offline.html")
        )
    );
});

self.addEventListener("fetch", e => {
    e.respondWith(
        fetch(e.request).catch(() =>
            caches.match("offline.html")
        )
    );
});
```

Manifest.json

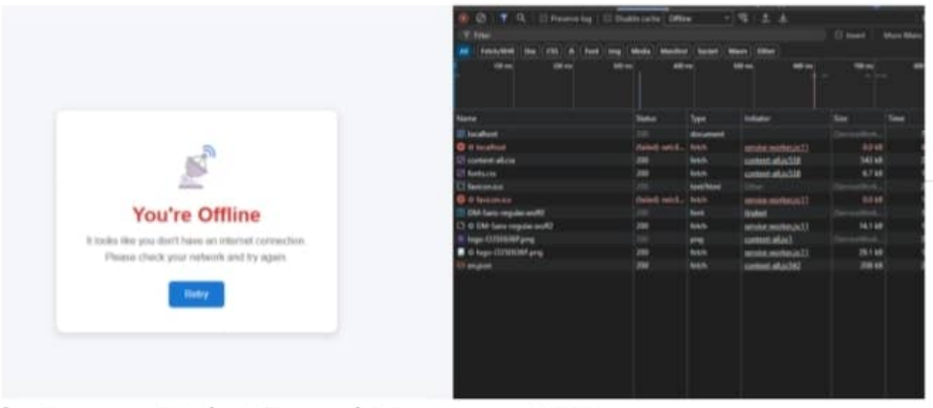
```
    "name": "College Information portal",
    "short_name": "CIP",
    "start_url": "/",
    "display": "standalone",
    "icons": [
      {
        "src": "/icons/icon1.png",
        "sizes": "512x512",
        "type": "image/png",
        "purpose": "any"
```

layout.cs.html

```
        if ('serviceWorker' in navigator)
        {
            navigator.serviceWorker.register('/servise-worker.js');
        }
    </script>
    <link rel="stylesheet" href="~/practical_7.styles.css" as
    <link rel="manifest" href="manifest.json"/>
</head>
```

Design Offline page :

Output :



b. Create a Student Record Management PWA.

Code :

```
const CACHE = "v1";

self.addEventListener("install", (e) => {
    e.waitUntil(
        caches.open(CACHE).then((cache) => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js",
                "offline.html"
            ]);
        })
    );
});

self.addEventListener("fetch", (e) => {
    e.respondWith(
        caches.match(e.request).then((response) => {
            return response || fetch(e.request);
        })
    );
});
```

student.cs | _Layout.cshtml | manifest.json | Index.cshtml | servise-worker.js

practical 7 | WebApplication8.Models.student

```
namespace WebApplication8.Models
{
    2 references
    public class student
    {
        1 reference
        public int RollNo { get; set; }
        1 reference
        public string Name { get; set; }
    }

}
```

student.cs | _Layout.cshtml | manifest.json | Index.cshtml | servise-worker.js | HomeController.cs

practical 7 | AspNetCoreGeneratedDocument.Views_Home_Index | ExecuteAsync()

```
@{
    ViewData["Title"] = "Home Page";
}

<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>

    <p>Roll No: @Model.RollNo</p>

    <p>Name: @Model.Name</p>
</div>
```

```
0 references
public class HomeController : Controller
{
    0 references
    public IActionResult Index()
    {
        student s = new student();

        s.RollNo = 04;
        s.Name = "Vijay Bisht";

        return View(s);
    }
```

Output :

practical_7

- Home
- Privacy

Welcome

Learn about building Web apps with ASP.NET Core .

Offline Page

Student Record

Roll No: 04

Name: Vijay Bisht

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